USB POWER & COMMUNICATION

VUSB MAX = 5V

HIGH = 2.31V - 3V3

V_IO = (150k/(100k + 150k)) x 5V = 3V (HIGH)

BATTERY CHARGING AND PROTECTION

Charging current - 250mA

502030 Battery capacity - 250mAh

Full charge time - 250mAh/250mA = 1hour

MCU

POWER MUX

MODE = VIN - Priority mode

Minimum VREF = 1V

VREF = (150k/(100k + 150k)) x 5V = 3V

3V3 BUCK-BOOST

MODE = 1 - Minimum output ripple

BUTTONS

BATTERY POWER

VBAT MAX = 3.7V

ESP32 VREF = 1.1V

VADC_MAX = (120k/(120k + 300k)) x 3.7V = 1.0571V

OLED

Sheet: /
File: list-tap.kicad_sch

Title: list-tap

Size: User
Date: 2026-10-19
Rev: 2nd
KiCad E.D.A. 10.0.5
Id: 1/1